

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. DATA TRANSMISSION PROTOCOLS :

Within the ComicCraft architecture, "Communication" denotes the continuous, asynchronous data exchange between the local FastAPI host and the remote Generative AI servers. Since the platform operates by chaining multiple AI models together, establishing highly secure and rapid communication channels was the highest development priority.

3. EXTERNAL API DIALOGUE

- **Narrative Processing (Google Gemini):** The backend interacts with Google's servers via RESTful APIs. It transmits the user's prompt as an HTTP POST payload. In return, the Gemini API responds with a structured JSON array (for the logical outline) and expansive text strings (for the dialogues).
- **Visual Asset Retrieval (Hugging Face):** The application communicates with the Stable Diffusion endpoint utilizing Python's `urllib.request`. It sends a JSON payload containing the scene's visual description along with a secure authorization token, receiving raw image byte data in return, which is then saved locally.

4. CLIENT-SERVER UI FEEDBACK:

Clear communication with the user is maintained via the frontend HTML/Jinja2 interface.

- **Status Indication:** Given the inherent latency of external AI generation, the system communicates processing times to users to prevent accidental form resubmissions.
- **Exception Communication:** In the event of an external API timeout, the backend intercepts the failure and communicates this to the frontend by

rendering a localized backup placeholder graphic, preventing a complete system crash and ensuring the user is aware of the network limitation.